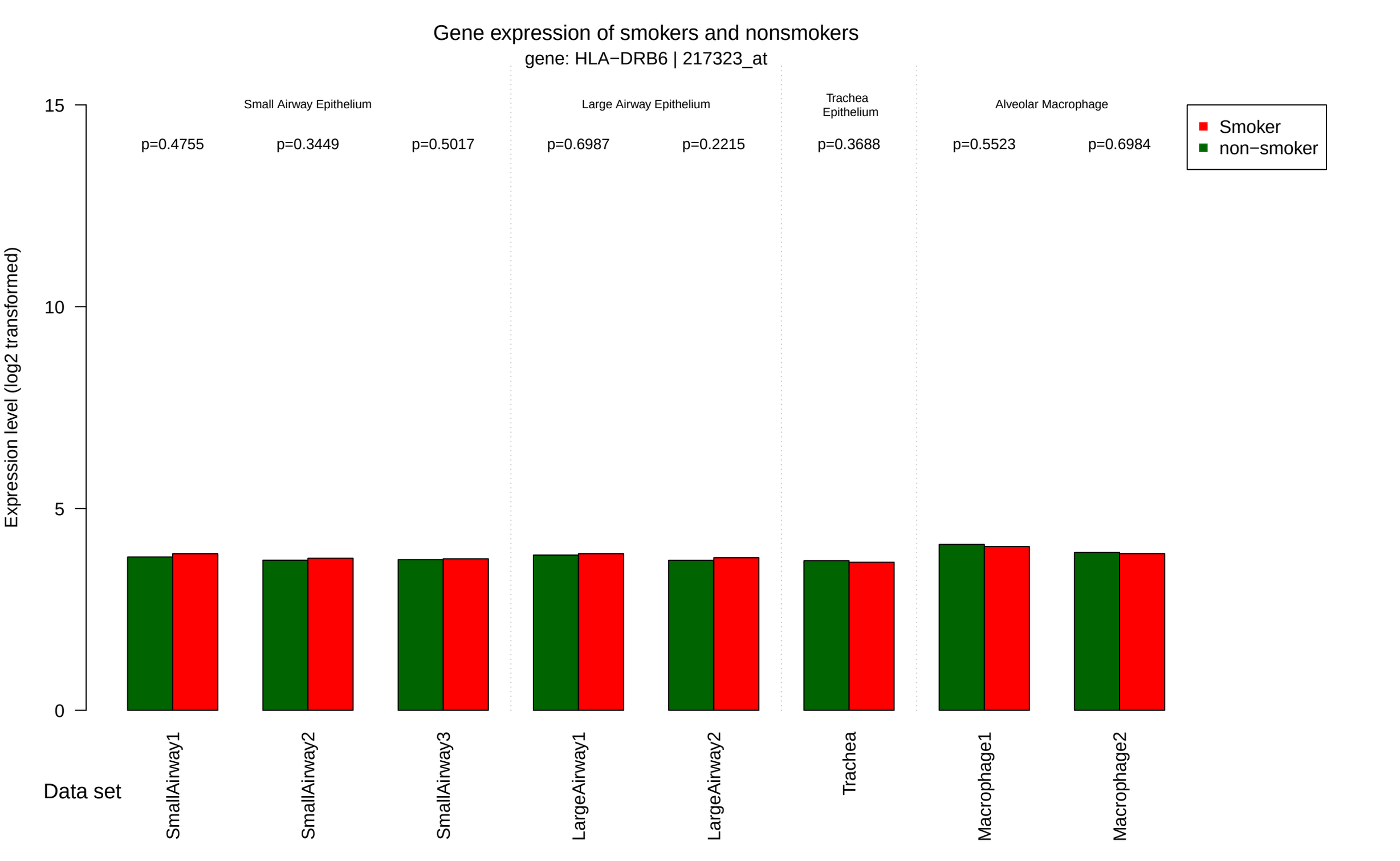
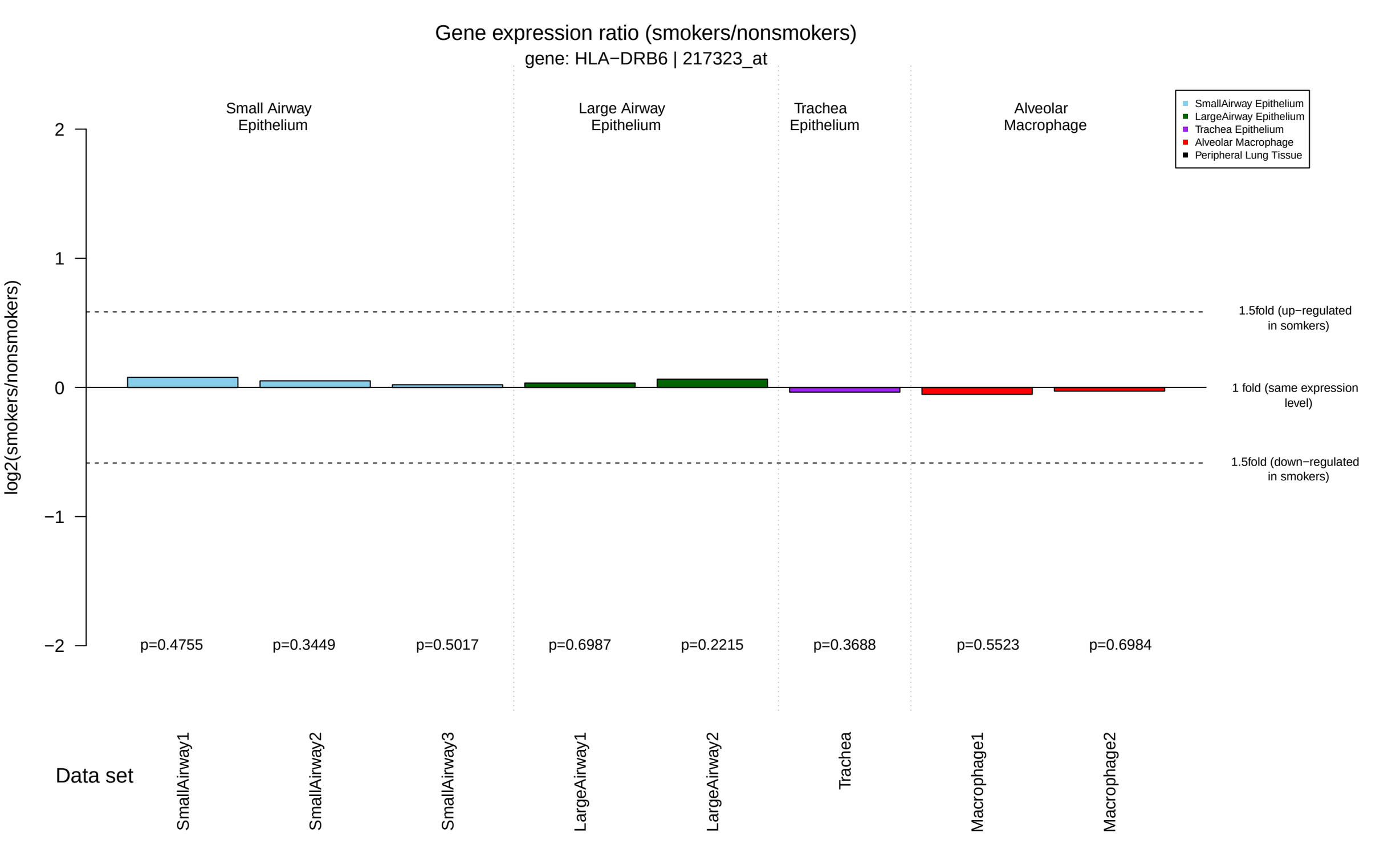
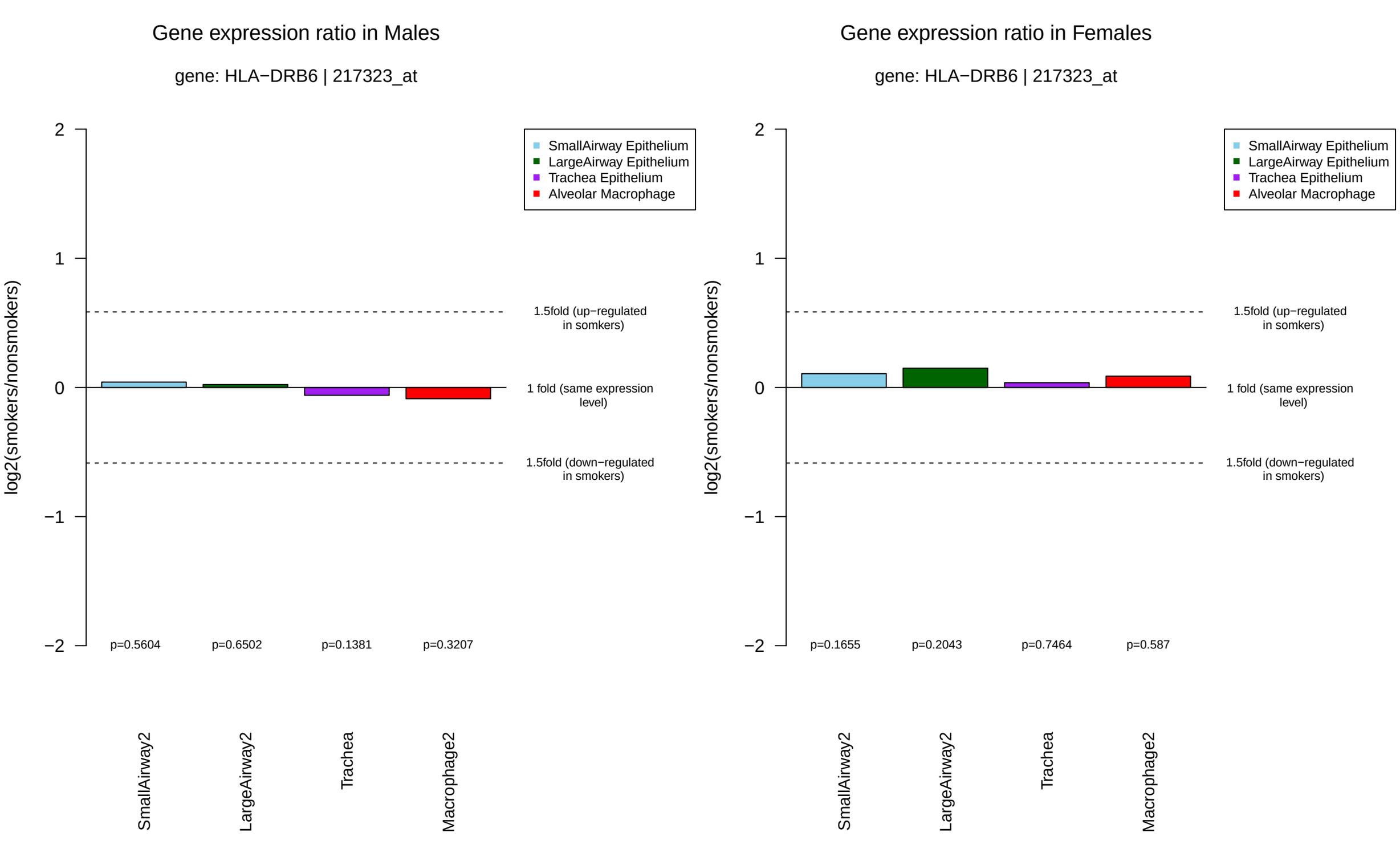
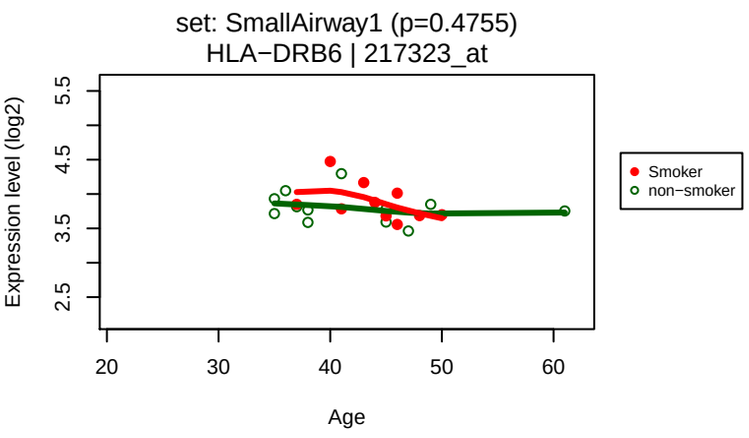
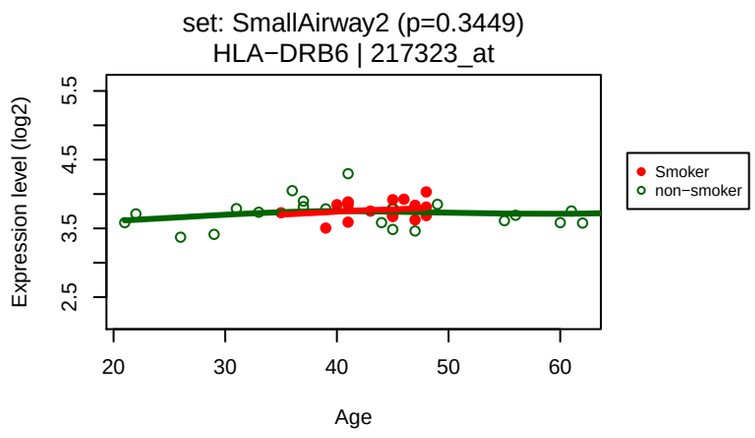
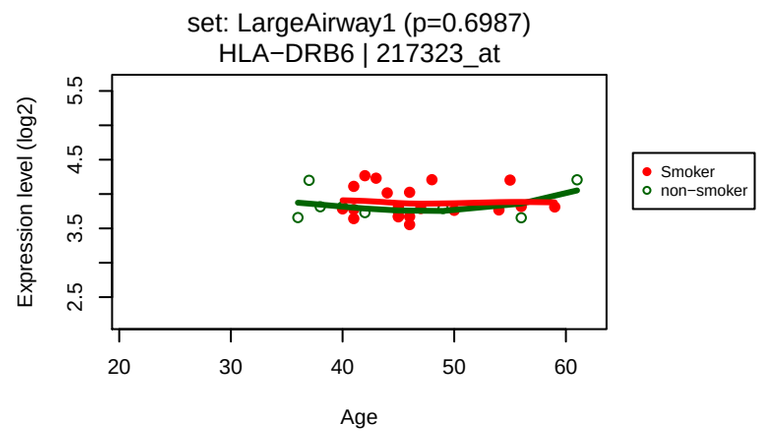

Gene expression of smokers and nonsmokers
gene: HLA−DRB6 | 217323_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.4755
p=0.3449
p=0.5017
p=0.6987
p=0.2215
p=0.3688
p=0.5523
p=0.6984
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

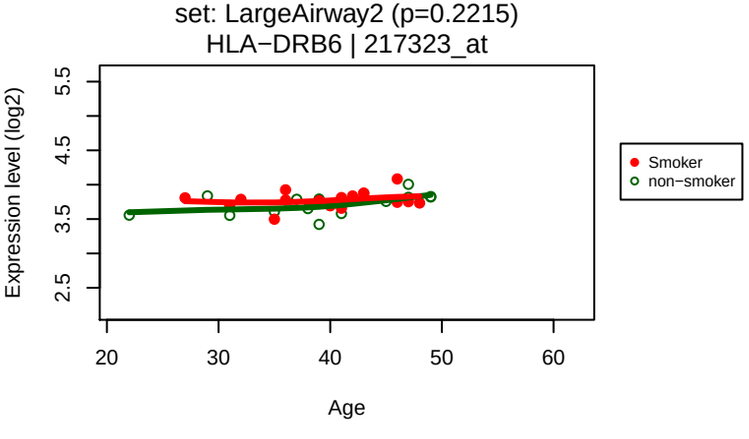

Gene expression ratio (smokers/nonsmokers)
gene: HLA−DRB6 | 217323_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4755
p=0.3449
p=0.5017
p=0.6987
p=0.2215
p=0.3688
p=0.5523
p=0.6984
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

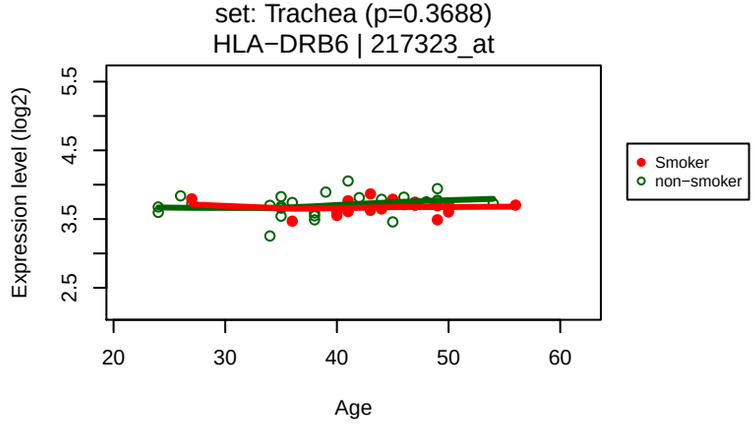

Gene expression ratio in Males
gene: HLA−DRB6 | 217323_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5604
p=0.6502
p=0.1381
p=0.3207
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: HLA−DRB6 | 217323_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1655
p=0.2043
p=0.7464
p=0.587
SmallAirway2
LargeAirway2
Trachea
Macrophage2

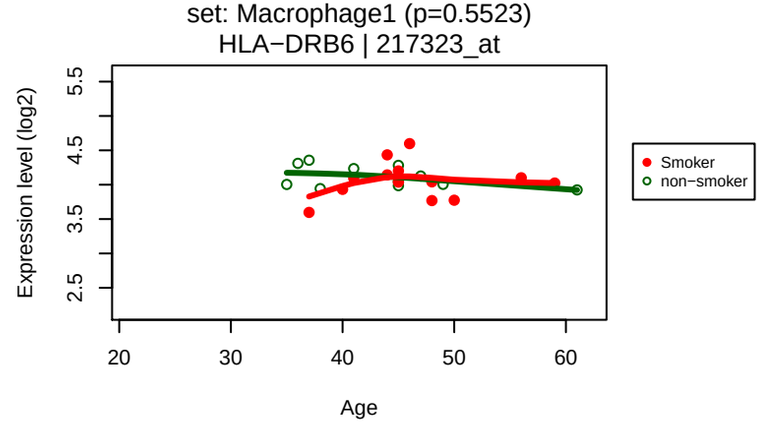

set: SmallAirway1 (p=0.4755)
HLA−DRB6 | 217323_at
Expression level (log2)
Age
Smoker
non−smoker

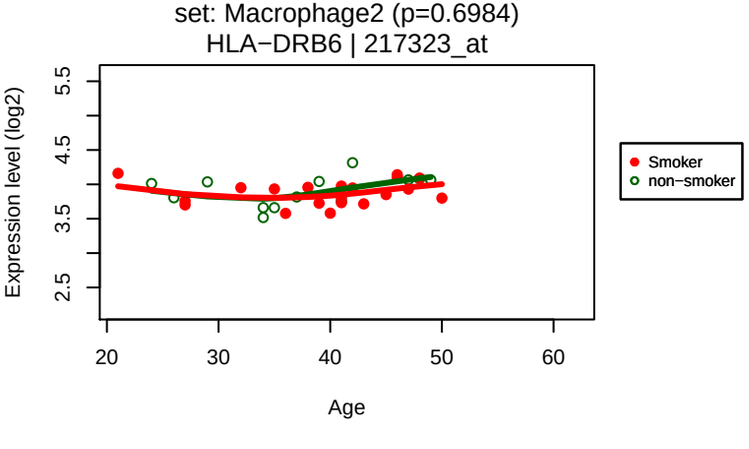

set: SmallAirway2 (p=0.3449)
HLA−DRB6 | 217323_at
Expression level (log2)
Age
Smoker
non−smoker

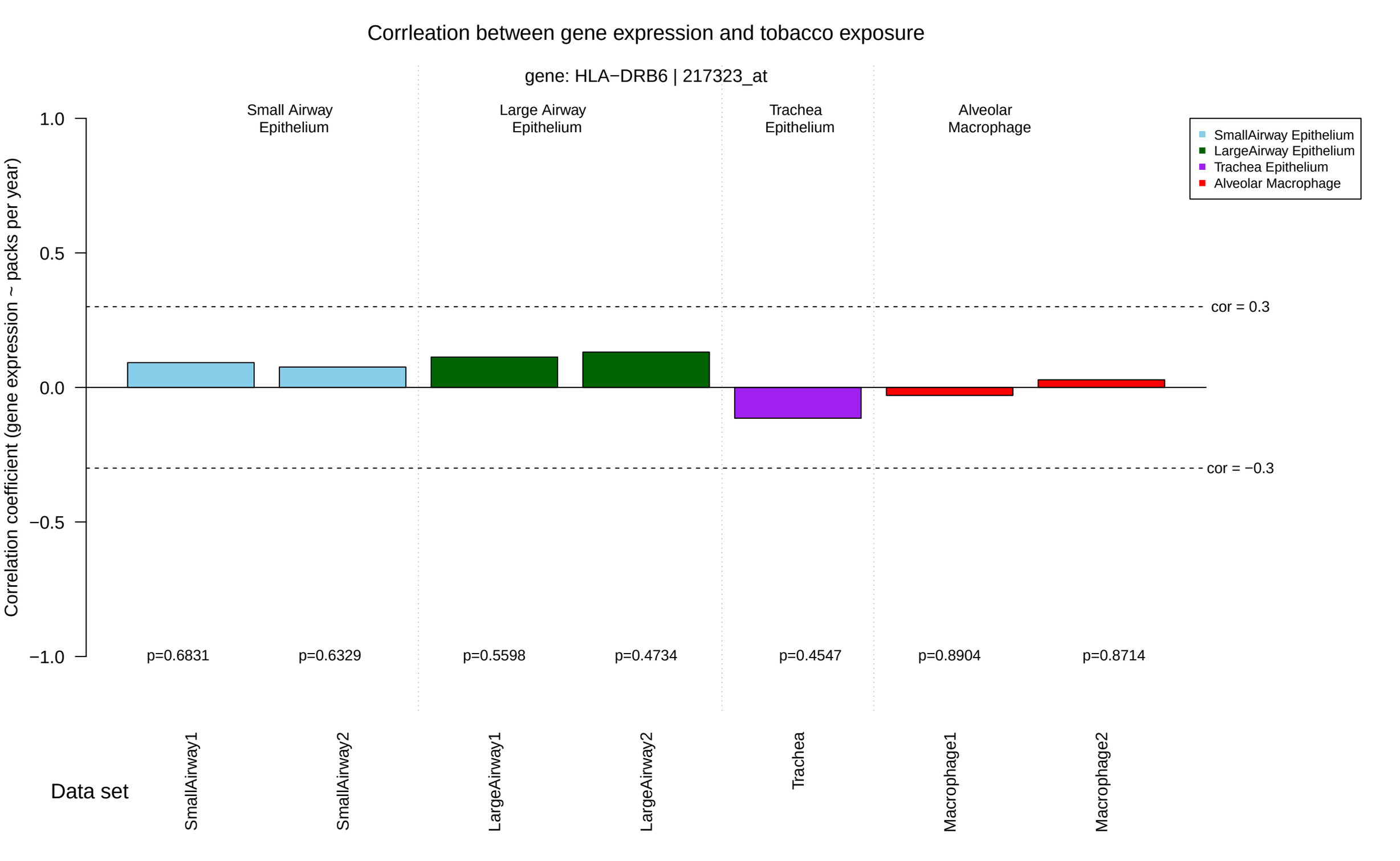

set: LargeAirway1 (p=0.6987)
HLA−DRB6 | 217323_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.2215)
HLA−DRB6 | 217323_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.3688)
HLA−DRB6 | 217323_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.5523)
HLA−DRB6 | 217323_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.6984)
HLA−DRB6 | 217323_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: HLA−DRB6 | 217323_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.6831
p=0.6329
p=0.5598
p=0.4734
p=0.4547
p=0.8904
p=0.8714
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2